Shubhankar Gupta

AI Engineer | LLM Agents (LangGraph, MCP), RAG, Evaluation, Python, Cloud Run

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SUMMARY

AI engineer who builds and runs LLM systems end to end: retrieval, tool servers, evaluation and production serving. Sole engineer behind **solquara.com**, a live AI video-ad studio with a generative-media pipeline and an accounts platform covered by **125 end-to-end checks**. Publishes open-source agent, retrieval and evaluation work with measured results: a LangGraph policy guard kept an eager agent at **40/40** compliant conversations (28/40 without it), and hybrid search in Postgres reached **nDCG@10 0.712 on BEIR SciFact**. Self-taught engineer with eight prior years running operations and client service for a family business.

SKILLS

Languages: Python, TypeScript, JavaScript (ES2022), SQL, PHP 8

LLMs and generative AI: Anthropic Claude API, OpenAI API, Gemini API, prompt engineering, tool and function calling, structured JSON output, multimodal generation (image, video, TTS)

RAG and search: Retrieval-augmented generation, embeddings, hybrid search (BM25 + dense vectors), reciprocal rank fusion, reranking, pgvector, HNSW, citation grounding

Evaluation: LLM and agent evaluation, retrieval metrics (Recall@k, MRR, nDCG), pass^k, Cohen's kappa, McNemar test, bootstrap confidence intervals

Agents: LangGraph (state graphs, checkpointing, human-in-the-loop interrupts), tool calling, policy guards, Model Context Protocol (MCP), JSON-RPC

Backend and data: FastAPI, Pydantic, Node.js, REST APIs, PostgreSQL, SQLite, MySQL, webhooks (HMAC), web crawling, data pipelines

Cloud and DevOps: Google Cloud Run, Terraform, Artifact Registry, Secret Manager, Workload Identity Federation, Docker, GitHub Actions CI/CD

Machine learning: scikit-learn, pandas, NumPy, NLP, classification, regression

Testing and delivery: pytest, Vitest, end-to-end testing, Git, performance profiling, WebGL

Currently learning: LangChain, LlamaIndex

EXPERIENCE

Founder and Lead Engineer, Solquara (solquara.com) **Jul 2026 – Present**

AI video-ad and automation studio, remote. Sole engineer: architecture, AI pipeline, testing, deployment and incident response.

- Designed the **generative-media pipeline** (keyframe image, then image-to-video, then voiceover). Costed the models first: for a 25-second Short, keyframe-then-animate was **~4x cheaper** than an all-in-one narrated-video workflow.
- Built the **accounts, plans and billing platform** (auth, 4 plan tiers, usage limits, signed billing webhooks with replay protection) and covered it with **125 end-to-end checks**, including IDOR, session and throttling tests, plus **112 checks against production**.
- Rebuilt the public site as **40 routes** in vanilla JavaScript and WebGL. Homepage went from 23 stylesheets to 2, 26 scripts to 1, and **123 kB to 36 kB** of HTML.
- Profiled a scroll-driven UI, found one CSS effect using **~80% of the frame budget**, and took it from **6 fps to 55 fps**.
- Built a Python **B2B prospecting pipeline** (OpenStreetMap extraction, resumable crawler, Companies House verification) that produced **~920 verified prospects**.
- Found a bug every automated test missed (a duplicate animation hid the whole pricing table) and added a build check that fails if any price is not rendered in the HTML.

Independent AI Engineer, Open-source projects **2024 – Present**

- Published the agent, voice, retrieval and evaluation projects below. Each has its own test suite and CI, and runs on a clean checkout with no API key.

Operations and Client Services, Family handicraft and handloom business, Delhi **2016 – 2024**

- Ran daily operations end to end (inventory, fulfilment, pricing, after-sales) without a supervisor. Handled **20+ client conversations a day** and used sales and stock data to decide what to buy.

PROJECTS

handoff-graph — Python, LangGraph, FastAPI, Claude

- Card-support agent in **LangGraph**: intent routing, an identity-verification gate, a policy guard before every tool call, and human handoff with interrupt() and resume. Graded on final account state plus an independent audit of the action log.
- Across 40 conversations, an eager planner stayed at **40/40** with the guard, but broke policy in **12 of 40** without the gate and guard. 15 tests; Docker image and API smoke-tested in CI.

agentcheck — Python, Anthropic SDK, LLM evaluation

- Evaluation harness for tool-using agents: **36 tasks** in a deterministic 7-tool environment, graded on final state, a policy check over the action log, and the answer.
- Showed a transcript-only grader agreed with ground truth **worse than chance (Cohen's kappa -0.40)** and passed 16 policy-breaking runs, and that a 96% pass@1 agent falls to **75% pass^8**. 22 tests; CI regenerates the report.

voice-latency — Python, Vosk streaming STT, TTS, voice activity detection

- Benchmarked every stage of a voice-agent turn offline on 48 clips in four English accents. A 300-500 ms silence threshold cut off **every** caller who paused mid-sentence; only 1200 ms caught them all, so endpointing, not the model, set the latency floor.
- Streaming recognition left only ~30 ms of work after the turn ended, but word error rate was **2.1% on the US voice vs ~11%** on UK, Australian and Indian English. 7 tests; CI runs a Linux smoke benchmark.

pg-hybrid-search — TypeScript, PostgreSQL, pgvector, Docker

- Hybrid search in one SQL query (BM25 + pgvector, fused with RRF), benchmarked on BEIR SciFact: **nDCG@10 0.712** vs 0.643 for vectors alone.
- Found pgvector's default ef_search silently returns **40 rows for LIMIT 100**. Tests run on PGlite and real Postgres in CI, which caught two portability bugs.

aistudio-mcp — Node.js, MCP, Gemini API

- Zero-dependency **MCP server** giving Claude 9 tools: Gemini text and vision, image generation, Veo 3.1 video, TTS and the Files API. Long renders resume instead of timing out.
- **12 end-to-end tests** run the real server against a local mock of Google's API; CI on Node 18, 20 and 22.

rag-document-qa — Python, FastAPI, Claude, Terraform, Cloud Run

- Citation-grounded document Q&A. Measured that a similarity threshold could not separate off-topic questions (0.907 vs 0.905) and replaced it with a coverage guard that can.
- Packaged for **Google Cloud Run with Terraform** (Artifact Registry, Secret Manager, keyless GitHub Actions deploys via Workload Identity Federation). Found and closed a **file-read vulnerability** in its own API while hardening it. 69 tests.

Also on GitHub: job-radar (collects ~12,500 job postings a run from 9 sources with an eligibility verdict for each; 67 tests), hybrid-rag-eval (BM25 beat hybrid retrieval on Recall@3 on a small corpus and ran 40x faster), promptdesk (support agent with RAG and sentiment-aware escalation).

Earlier ML projects: heart disease classification (88.5% test accuracy, recall 0.92), wine quality prediction (13 models; 93% accuracy, 0.95 ROC-AUC), content-based movie recommender (TMDB 5000).

EDUCATION

Bachelor of Arts, Tourism, Indira Gandhi National Open University (IGNOU) — **2020**

CBSE Class XII, Air Force Bal Bharati School, New Delhi — **2014**

Engineering is self-taught; the evidence is public at github.com/shubhankar360.

LANGUAGES

English (fluent, professional), Hindi (native)